

Chapter 1 - Marked Thesis Revision

LEAPRS | Original chapter order and narrative retained | System details aligned with the system concept skill

Changed wording

New wording

Removed wording

The red text is retained for comparison, followed by its yellow replacement. Green marks added headings. University procedures and the pilot sample require local confirmation.

Original line 3

The Problem and Its Background

Original line 4

~~Training and professional development are core responsibilities of any university seeking to keep its workforce competent and updated, and meeting this responsibility depends on having an administrative system capable of handling the planning, budgeting, approval, and monitoring of training activities in a consistent and organized way. Wibawa (2024) found that incorporating digital technologies into administrative processes improves productivity, simplifies operations, and supports better informed decision making through more accurate data and streamlined resource management. Kayanja et al. (2025) added that in higher education specifically, this shift extends beyond teaching and learning into the administrative side of running a university, as manual systems eventually become harder to sustain as institutional needs grow. These findings frame the administrative challenge faced by Pamantasan ng Lungsod ng Pasig in managing its own training requisition process.~~

Training and professional development are core responsibilities of a university seeking to keep its teaching and non-teaching workforce prepared for changing institutional needs. Meeting this responsibility requires an administrative process that can organize requests, supporting documents, budgets, and progress records. In a bibliometric study of digital transformation and administrative efficiency, Wibawa (2024) discusses the potential of digital tools to support more organized operations and informed decisions. Kayanja et al. (2025), studying the move toward paperless systems in higher education, describe both opportunities and practical challenges in such a transition. These studies frame the administrative problem faced by Pamantasan ng Lungsod ng Pasig as it manages training requisitions through records that are still partly physical and dispersed.

Original line 5

Every school year, Pamantasan ng Lungsod ng Pasig conducts a structured capacity development program for its teaching and non-teaching personnel, guided by the university's Strategic Plan and CAPDEV. This program is managed by the Office of Lifelong Learning, headed by the Lifelong Learning Head, who reviews, approves, and oversees all training requests submitted across the university.

Original line 6

Under the current process, teaching and non-teaching personnel prepare physical documents to apply for a training activity. These documents pass through multiple offices for review and signature before final approval. Once submitted, the requestor does not report back on the request's status, and no record shows which office currently holds it or how long it has stayed there. As a result, the Lifelong Learning Head must physically visit each office to get updates.

Original line 7

Document review is similarly affected. Before approving a request, the Lifelong Learning Head must confirm that supporting documents are complete and that the associated cost does not exceed the

allocated budget. This can currently only be done by physically reviewing hard copies once they reach the Lifelong Learning Head's office, requiring the requestor to prepare and submit physical documents first before any of this can be checked.

Original line 8

The same visibility gap affects the Lifelong Learning Head's other responsibilities. Budget utilization from the CAPDEV is tracked through manually updated spreadsheets, and post-training document compliance is checked one physical submission at a time, both of which can only be reviewed by manually going through records rather than through a consolidated system.

Original line 9

~~LEAPRS is designed to address this gap by giving the Lifelong Learning Head a way to monitor each request's status throughout its progression. Requestors upload their supporting documents through the system first, allowing the Lifelong Learning Head to check completeness and budget sufficiency digitally before approving the request to proceed with the corresponding physical process. As a request moves through the approval chain, its status is recorded and updated within LEAPRS, and the system notifies the Lifelong Learning Head of any change, removing the need for in-person follow-up just to ask what has happened. The same visibility extends to budget utilization and post-training compliance through a single dashboard.~~

LEAPRS is designed to address this gap by giving the Lifelong Learning Head a way to monitor each request's recorded progress without relying solely on in-person follow-ups. Requestors can submit requests and supporting documents through the system, allowing authorized staff to review the information and the remaining budget of the associated CapDev project before a request proceeds through the university's required physical process. As personnel report developments, they can add updates to the request's timeline; involved users receive in-app notifications of relevant changes. The system also places CapDev budgets, request histories, and submitted documents in a shared workspace. LEAPRS records the status supplied by its users, while the actual movement and approval of physical documents remain governed by the university's offices.

Original line 10

~~LEAPRS also includes a post-seminar evaluation feature. After each training activity, requestors complete a satisfaction survey and a short, open-ended learning survey. Scale-based responses are processed statistically, while free-text responses are analyzed through Google's Gemini API, with HDBSCAN grouping related responses into themes for higher-volume submissions, producing thematic summaries the Lifelong Learning Head can review without reading every response individually.~~

LEAPRS also supports evaluation after a training activity has been marked complete. It creates a participant evaluation form and a supervisor or post-activity evaluation form for each completed request. The forms collect scale-based ratings and written feedback. LEAPRS calculates distributions and averages from the ratings, while Groq, when configured, summarizes written comments into strengths, areas for improvement, and recommended actions. These summaries help the Lifelong Learning Head review feedback alongside the original responses; they do not replace the Head's judgment or constitute a validated thematic analysis.

Original line 11

~~Existing literature addresses document tracking, administrative digitization, and AI-assisted text analysis separately, but none of it integrates these functions into a single platform built for the tracking~~

~~and monitoring needs of an office like the Lifelong Learning Head's. This gap forms the basis for the present study.~~

Existing literature addresses personnel development, document tracking, administrative digitization, and AI-assisted analysis of written feedback. The present study brings these concerns together in the particular setting of PLP's capacity development and training requisition process. Rather than claiming that no other institution has combined them, the study examines whether LEAPRS provides an appropriate and usable response to the local problems identified with the Lifelong Learning Office.

Original line 12

Review of Related Literature

Original line 13

Purpose of Training and Professional Development for Teaching and Non-Teaching Personnel

Original line 14

~~Training programs in higher education serve different purposes depending on the personnel they are designed for. Fedeli et al. (2025) describe faculty development programs as instruments meant to advance instructional effectiveness, course design, and pedagogical innovation, with the goal of helping faculty apply new teaching competencies in their actual classroom practice. describes faculty development programs as instruments meant to advance instructional effectiveness, course design, and pedagogical innovation, with the goal of helping faculty apply new teaching competencies in their actual classroom practice. This shows that training for teaching staff is closely tied to improving the quality of instruction delivered to students, making it a core investment area for any higher education institution such as PLP.~~

Training programs in higher education serve different purposes depending on the personnel they are designed for. Fedeli et al. (2025), in their discussion of the Teaching4Learning program at the University of Padova, connect faculty development with active learning, instructional innovation, and the growth of a teaching community. Their work shows that professional development for teaching staff is linked not only to attendance at training but also to how faculty apply new approaches in their academic practice. This makes the planning and documentation of faculty development relevant to institutions such as PLP.

Original line 15

~~Non-teaching personnel, on the other hand, require training with a different focus. Alonzo (2025) conducted a learning needs analysis of non-teaching personnel at a Philippine state college and found that staff across different roles, including supervisors, clerical workers, and utility personnel, had distinct skill gaps ranging from planning and leadership to technical and administrative functions. The study recommended that institutions design role-specific, targeted training programs rather than applying a single generic training approach to all non-teaching staff. This distinction is directly relevant to LEAPRS, since the system's CAPDEV entries and training requisition records need to account for the fact that faculty and non-teaching personnel pursue training for different purposes and toward different competencies, which affects how training activities are categorized, tracked, and evaluated within the platform.~~

Non-teaching personnel, on the other hand, require development that reflects their own responsibilities. Alonzo (2025) examined the learning needs of non-teaching personnel at Bulacan Agricultural State College and found differences in skills and training needs across roles. The study

supports a targeted approach instead of treating all personnel as a single training group. This distinction is relevant to LEAPRS because its CapDev records and configurable request forms can hold information about different activities and participants, even though the system does not automatically classify competencies by personnel type.

Original line 16

Digital Transformation in Administrative Management

Original line 17

~~Organizations across different sectors have moved from manual to digital systems as their operations grow more complex. Wibawa (2024) found that incorporating digital technologies into administrative processes improves productivity, simplifies operations, and supports better informed decision-making through more accurate data and streamlined resource management. Kayanja et al. (2025) added that in higher education specifically, this shift extends beyond teaching and learning into the administrative side of running a university, as manual systems eventually become harder to sustain as institutional needs grow. These findings form the basis of the argument that PLP's Lifelong Office, which still relies on paperwork and spreadsheets to manage training requisitions, needs a digital system to keep up with the growing demands of the university's capacity development program.~~

Organizations across different sectors have moved from manual to digital administrative systems as the amount of information they handle grows. Wibawa (2024) reviewed work on digital transformation and administrative efficiency, emphasizing the role of data and digital tools in organizing operations and supporting decisions. In higher education, Kayanja et al. (2025) examined the move toward automated and paperless systems at Makerere University Business School and identified both potential benefits and barriers, including digital skills and resource constraints. Their findings suggest that PLP's shift from paper-based training requisition records to LEAPRS must consider not only the technology but also the people and processes that will use it.

Original line 18

Synthesis

Original line 19

~~Together, these four sources establish the foundation for LEAPRS' core rationale. Fedeli et al. (2025) describe faculty development programs as instruments meant to advance instructional effectiveness, course design, and pedagogical innovation, with the goal of helping faculty apply new teaching competencies in their actual classroom practice. Alonzo (2025) shows that teaching and non-teaching personnel pursue training for different purposes and toward different competencies, which grounds why LEAPRS' CAPDEV entries and training requisition records need to account for these distinctions rather than treating all training as identical. Wibawa (2024) and Kayanja et al. (2025) establish that digital transformation in administrative work is a requirement rather than an option for institutions facing growing operational demands, which reflects PLP's current situation. Taken together, these four sources justify both the need for LEAPRS as a digital solution and the requirement that it treat teaching and non-teaching personnel as distinct user groups with distinct training needs.~~

Taken together, these four sources establish the rationale for a system that can record and monitor varied personnel development activities. Fedeli et al. (2025) place faculty development in the context of improving teaching practice, while Alonzo (2025) shows why non-teaching personnel's training needs also deserve attention. Wibawa (2024) and Kayanja et al. (2025) explain the administrative value and practical challenges of digital transformation. For PLP, these studies support an organized

digital record of CapDev projects and requests, while the suitability of LEAPRS for its users must still be evaluated in the local setting.

Original line 20

Web-Based Tracking and Monitoring Systems in Philippine Higher Education

Original line 24

~~Document tracking has long been a challenge for administrative offices in Philippine state universities and colleges. Piloton et al. (2023) developed a records tracking management system using QR codes for a Philippine state college, designed to help the institution monitor, record, and track physical documents as they move between offices. Their evaluation involved office personnel who assessed the system's usability and effectiveness in reducing manual tracking effort. This finding supports the direction of LEAPRS, since the system also aims to give the Lifelong Head a way to track training requisition documents from submission to approval instead of relying on physical folders and manual logs. Since Piloton et al.'s study was also set in a Philippine SUC, it reflects the same institutional environment PLP operates in, strengthening the case for a similar tracking approach applied specifically to training requisition management.~~

Document tracking has long been a challenge for administrative offices in Philippine higher education. Piloton et al. (2023) developed a QR-code records tracking system at Surigao del Norte State University to record the movement of physical documents that had previously been entered in a logbook. Their work shows the importance of making a record's progress visible to the personnel responsible for it. LEAPRS responds to a related problem in training requisitions, but through digitally submitted requests and user-entered status timelines rather than QR-based tracking of a physical folder.

Original line 22

Web-Based Systems for Workflow, Document Management, and Budget Monitoring

Original line 23

~~Web-based platforms are widely used to manage administrative workflows and reduce the need for physical processes. Widiyatmoko et al. (2024) built a web-based registration system and found that digitizing submission, document upload, and status tracking improved both efficiency and accessibility compared to manual processes. This matches what LEAPRS aims to do for training requisition submission, since employees will be able to submit requests and monitor status online instead of relying on paper forms.~~

Web-based platforms can reduce dependence on paper-based submission and scattered records. Widiyatmoko et al. (2024) developed a web-based student registration information system using a Rapid Application Development approach and reported that the application met their testing criteria. Although student registration differs from training requisition, the study provides an example of using a web application to organize an administrative submission process. LEAPRS applies this general approach to employees' CapDev-related requests, supporting documents, and recorded status updates.

Original line 24

~~Afandi and Nuralmasari (2026) studied a web-based document governance system for a government agency and found that centralizing document workflows through one platform reduced the risk of lost files and supported real-time compliance checking. This reflects the same document management problem the Lifelong Head currently faces without an organized digital filing system. Madyatmadja et~~

al. (2022) showed that real-time dashboards give administrators a continuous view of budget use and activity timelines, which allows for quicker and more accurate decisions. This directly supports the budget monitoring feature planned for LEAPRS, since the Lifelong Head currently tracks training budgets manually with no live visibility into fund utilization.

Afandi and Nurmalasari (2026) examined a web-based document governance system for Tegal City's public information office, where manual collection across agencies had made document monitoring difficult. Their study is relevant to the value of a centralized view of submitted records. In LEAPRS, uploaded files are stored through Google Drive and linked to the corresponding request or status update, while CapDev records show the budget remaining after recorded deductions. These functions assist review, but LEAPRS does not automatically certify that every required post-training document has been submitted.

Original line 25

Digital Administrative Systems in the Philippine Public Sector

Original line 26

The relevance of digital systems is well-established in the Philippine local government context. Pajenado and Rivera (2025) evaluated the implementation of a digital permit system in a Philippine city government and found that it reduced processing time and improved transparency compared to the manual process it replaced, although adoption still faced some resistance due to unfamiliarity with the new system. Since PLP is a public university under the Pasig city government, it shares the same structural setup, including reliance on paperwork and limited visibility into ongoing transactions, as the local government unit in this study. This strengthens the argument that LEAPRS is suited for the specific operating conditions of a Philippine public institution like PLP, while also suggesting that user adoption support should be considered during LEAPRS' rollout.

Digital public services in the Philippines also illustrate the importance of adoption, not only technical availability. Pajenado and Rivera (2025) studied the Paspas Permit system in Valenzuela City and reported gains in convenience alongside barriers related to digital literacy, complexity, and awareness. PLP is a different institution with a different requisition process, so the study cannot establish LEAPRS's results in advance. It does, however, suggest that usability and orientation should be considered when introducing a digital workflow to personnel accustomed to physical submissions.

Original line 27

Synthesis

Original line 28

These five sources collectively support the specific system functions LEAPRS is being built to handle. Piloton et al. (2023) show that document tracking systems have already proven effective in a Philippine SUC setting, which supports the tracking function planned for LEAPRS. Widiyatmoko et al. (2024), Afandi and Nurmalasari (2026), and Madyatmadja et al. (2022) show that web-based platforms can manage submission, document compliance, and budget visibility more effectively than manual processes, the same functions LEAPRS is being built to handle. Pajenado and Rivera (2025) ground all of this in the local Philippine public sector context, confirming that digital systems produce measurable improvements in institutions similar to PLP. Together, these sources validate that LEAPRS' core requisition, document, and budget-tracking features are grounded in proven, real-world implementations rather than untested assumptions.

These studies collectively support the functions around which LEAPRS is organized. Piloton et al. (2023) show the value of traceable records in a Philippine university setting; Widiyatmoko et al. (2024) provide an example of web-based administrative submission; Afandi and Nurmalasari (2026) address centralized document monitoring; and Pajenado and Rivera (2025) show that user adoption matters in Philippine public services. Their settings and methods differ from PLP's. Accordingly, they provide a foundation for the proposed system, while its effects on requisition monitoring, budget visibility, and user experience remain questions for this study to examine.

Original line 20

~~Large Language Models for Thematic Analysis of Open-Ended Survey Text~~

Large Language Models for Summarizing Open-Ended Evaluation Feedback

Original line 30

~~Researchers have increasingly applied large language models to the analysis of unstructured, open-ended text. De Paoli (2024) built on earlier work applying large language models to thematic analysis, proposing prompts that allow a model to perform initial coding and theme generation directly from open-ended qualitative data, then evaluating the resulting themes against those produced manually by human researchers. This supports the feasibility of using a large language model such as Gemini to generate theme labels from raw survey text without requiring a human coder to read every response individually, which is the same free-text challenge the Lifelong Learning Head currently faces when reviewing post-seminar learning surveys manually.~~

Researchers have increasingly explored large language models as aids for reading open-ended text. De Paoli (2024) proposed prompts for initial coding and theme development in qualitative material, illustrating how model output can help a researcher identify patterns. The study also makes clear that an AI-supported reading process depends on its prompts and the researcher's interpretation. For LEAPRS, this literature supports the narrower task of summarizing written evaluation comments for review; it does not establish that an automatically produced summary is a complete thematic analysis of employee learning.

Original line 34

~~Wongvorachan and Bulut (2025) closer to LEAPRS' actual use case, demonstrated the application of NLP techniques, including text preprocessing, topic modeling, and text summarization, directly on students' open-ended course evaluation survey comments, showing that written feedback in an educational setting can be systematically processed into themes and sentiment patterns instead of read manually one response at a time. Since a post-seminar learning survey is functionally the same kind of instrument as a course evaluation survey, both being open-ended feedback collected after a learning activity, this directly supports the feasibility of applying the same category of text analysis to LEAPRS' post-seminar learning survey responses.~~

Wongvorachan and Bulut (2026) discuss natural language processing in learning analytics and demonstrate techniques, including text preprocessing, topic modeling, and summarization, using student comments from course evaluations. Their educational setting makes the chapter relevant to LEAPRS's post-activity feedback, because both involve written responses about a learning experience. At the same time, their demonstration is not an evaluation of the LEAPRS forms or its Groq-generated summaries. It supports the feasibility of examining comments systematically, while the usefulness of LEAPRS's output must be judged with its intended users.

Original line 32

Katz et al. (2024) for larger volumes of open-ended data, proposed the Generative AI-enabled Theme Organization and Structuring (GATOS) workflow, which combines generative language models with machine learning techniques to approximate thematic analysis at scale. In this workflow, individual responses are first condensed into short summary points using a language model, the summary points are then converted into text embeddings, and those embeddings are grouped through dimensionality reduction and clustering so that semantically similar responses are organized together before a generative model labels each resulting cluster. This directly parallels LEAPRS' design for large-volume post-seminar responses, where responses are embedded and clustered before Gemini generates a label and summary for each cluster, while smaller volumes are processed directly, without a clustering step, consistent with De Paoli's approach.

Katz et al. (2026) present the GATOS workflow, which combines generative text models with other tools to develop qualitative codebooks from larger bodies of text. Their work demonstrates one possible approach to organizing extensive qualitative data, but LEAPRS does not implement GATOS, response embeddings, or clustering. The study remains relevant as a comparison point: it shows how more elaborate research-grade analysis differs from LEAPRS's concise, on-demand summaries of comments. Keeping this distinction prevents the system's practical feedback aid from being described as an unimplemented large-scale thematic pipeline.

Original line 33

~~Density-Based Clustering for Unsupervised Theme Discovery~~

Interpreting AI-Assisted Evaluation Summaries

Original line 34

Selecting an appropriate clustering algorithm is central to grouping open-ended responses into themes when the number of themes cannot be known in advance. Wani (2024) provides a comprehensive comparative survey of clustering methodologies and situates density-based approaches, including DBSCAN, OPTICS, and HDBSCAN, against centroid-based methods such as K-Means. The survey confirms that density-based methods group closely packed points into clusters while marking sparse regions as noise, without requiring the number of clusters to be fixed beforehand, a property that makes this family fundamentally better suited to LEAPRS' use case than K-Means, which requires the cluster count to be set in advance and is therefore incompatible with a system where the number of themes varies from one seminar's responses to the next.

Scale-based and open-ended responses contribute different kinds of information to an evaluation. In LEAPRS, rating distributions and averages come directly from submitted responses and can be inspected as quantitative summaries. Written comments require interpretation. The system uses Groq, when configured, to condense those comments into strengths, areas for improvement, and recommended actions, while preserving access to the original responses for human review.

Original line 35

Wani (2024) further notes that OPTICS was developed to address DBSCAN's reliance on a single global distance threshold by instead constructing a reachability-based ordering of points, which adapts to local variation in density rather than applying one fixed setting across an entire dataset. HDBSCAN extends this further through a hierarchical approach that removes the need to specify a distance threshold entirely, instead selecting the most stable groupings across a range of density levels. This directly informs LEAPRS' selection of HDBSCAN over standard DBSCAN, since post-seminar

~~responses across different themes cannot be assumed to form clusters of uniform density, and Wani's synthesis of the trade-offs among these three algorithms gives an up-to-date, independent basis for that choice.~~

This division matters because a concise AI summary may help a busy reviewer notice recurring concerns, but it can also omit minority views or flatten the context of a particular response. De Paoli (2024) demonstrates the importance of prompts and interpretation in AI-assisted qualitative work. Likewise, the distinction between the methods demonstrated by Wongvorachan and Bulut (2026) and the more extensive codebook workflow of Katz et al. (2026) shows that the analytical claim must match the method actually used. LEAPRS therefore presents its generated text as a review aid rather than a final research finding about training effectiveness.

Original line 36

~~Herrmann et al. (2024) supported the decision to reduce response embeddings to a lower-dimensional space before clustering, who empirically demonstrate that projecting high-dimensional embeddings into a topologically-informed lower-dimensional space before applying a density-based clustering algorithm substantially improves cluster separability and reduces the algorithm's sensitivity to its distance parameter. Their experiments show that this pairing does not merely make clustering more convenient but measurably improves performance on both synthetic and real datasets, including cases with clusters of uneven density, the same condition LEAPRS' post-seminar responses are expected to exhibit. This provides direct empirical grounding for LEAPRS' pipeline of embedding open-ended responses, reducing their dimensionality, and then applying HDBSCAN, rather than clustering the raw high-dimensional embeddings directly.~~

For the Lifelong Learning Head, the practical question is whether the combined rating charts, original comments, and generated summary make post-activity feedback easier to inspect. LEAPRS retrieves the latest Google Forms responses when the See Summary action is used. If AI summarization is unavailable, the rating charts can still be viewed. This design allows the study to evaluate the usefulness and clarity of the feedback display without claiming that a specific clustering algorithm or a validated theme-discovery pipeline has been implemented.

Original line 37

Synthesis

Synthesis

Original line 38

~~These five sources establish the technical foundation for LEAPRS' AI-assisted post-seminar evaluation feature. Wani (2024) establishes that density-based clustering algorithms, HDBSCAN in particular, can group data into an unknown number of clusters without requiring that number to be specified in advance. Herrmann et al. (2024) provide empirical support for pairing such algorithms with prior dimensionality reduction, together directly supporting LEAPRS' use of HDBSCAN on dimensionality-reduced embeddings to organize open-ended responses into themes before Gemini generates each theme's summary. De Paoli (2024), Wongvorachan and Bulut (2025), and Katz et al. (2024) confirm that large language models can perform thematic analysis on open-ended text at both small and large scale, directly supporting LEAPRS' use of Gemini to process free-text post-seminar learning survey responses without requiring the Lifelong Head to read every response manually. None of these five sources on their own cover the complete scope of what LEAPRS aims to do, but combined, they support every core function planned for the system.~~

De Paoli (2024) illustrates the possibilities and interpretive demands of language-model-assisted analysis, Wongvorachan and Bulut (2026) show that educational feedback can be examined with text-analysis methods, and Katz et al. (2026) describe a more elaborate codebook workflow for large qualitative datasets. Together, these sources justify studying AI-assisted review of written feedback, but they do not by themselves validate every summary the system produces. LEAPRS combines directly calculated rating summaries with Groq-generated summaries of comments for human review. This narrower, implemented approach fits the administrative purpose of the project and gives the evaluation a clear feature to assess.

Original line 39

Research Gap

Original line 40

~~Based on the literature reviewed, there are already studies on the training needs of teaching and non-teaching personnel, digital transformation in administrative management, document tracking, web-based workflow systems, digital administrative systems for the Philippine public sector, density-based clustering algorithms, and large language model applications for thematic text analysis. However, none of these studies focus specifically on a system built for managing training requisitions and monitoring a university's capacity development program under a Lifelong Office. Most of the systems found in the literature handle only one function at a time, either document tracking, budget monitoring, or open-ended response analysis, but not all of them together in a single platform made for training requisition management. LEAPRS addresses this gap by combining requisition submission, budget encoding, approval routing, post-training compliance, and AI-assisted thematic analysis of post-seminar survey responses into one system built specifically for the operations of the Lifelong Head at Pamantasan ng Lungsod ng Pasig. Additionally, while prior work applies large language models to thematic analysis of open-ended text, it does not evaluate which density-based clustering algorithm is best suited to organizing that text into themes when the number of themes is unknown, a gap this study addresses through the empirical comparison of DBSCAN, HDBSCAN, and OPTICS described in Chapter 3.~~

Based on the literature reviewed, there are studies on staff development, administrative digitization, document tracking, web-based submission, and analysis of open-ended text. However, the reviewed work does not directly address the combination of training requisition records, CapDev budgets, request timelines, supporting files, and post-activity feedback in PLP's Lifelong Learning Office. LEAPRS addresses this local design need by bringing these functions into one web-based system. This study examines the system's functionality and the experience of its intended users; it does not claim to empirically compare DBSCAN, HDBSCAN, and OPTICS, because that comparison is not part of the implemented LEAPRS workflow.

Significance of the Study

Original line 43

The development of LEAPRS is expected to benefit the following:

Original line 44

~~Pamantasan ng Lungsod ng Pasig. The absence of a centralized system for managing training requisitions has long placed unnecessary strain on the university's administrative operations. LEAPRS directly addresses this by digitizing the full requisition cycle, budget encoding, approval routing, post-~~

training document compliance, and budget utilization monitoring under a single, organized platform that improves both institutional efficiency and accountability.

Pamantasan ng Lungsod ng Pasig. LEAPRS can provide the university with a centralized record of CapDev projects, training requests, supporting documents, status histories, and remaining budgets. Its potential contribution is more consistent access to requisition information across authorized users. Whether that access improves efficiency or accountability at PLP will be determined through the study's evaluation rather than assumed from the system's availability alone.

Original line 45

~~The Lifelong Head. The manual oversight of training requests, budget entries, compliance requirements, and seminar rotation fairness across the university is a considerable and growing responsibility. LEAPRS consolidates these functions digitally, providing real-time approval tracking, notifications, automatic Google Drive organization of all teaching and NTP submitted documents, a complete audit trail, and AI-driven insights equipping the Lifelong Head with the tools necessary to manage the program with greater precision and less administrative workload.~~

The Lifelong Learning Head. Oversight of training requests, supporting documents, and budget entries requires the Head to bring together information that is currently handled through separate physical and spreadsheet records. LEAPRS is designed to place recorded request progress, CapDev balances, relevant in-app notifications, audit history, and post-activity feedback summaries within reach of the authorized administrator. This may reduce the need for routine in-person status inquiries while preserving the Head's responsibility to review and decide on the information presented.

Original line 46

~~Teaching and Non-Teaching Personnel. Submitting a training requisition currently requires physical document preparation and in-person follow-ups across multiple offices. With LEAPRS, requestors can first upload their requisition documents through the system for the Lifelong Head's review before proceeding with the corresponding physical, hard-copy process required by each approving office. As the request moves through the approval chain, its status is tracked and updated within the system, allowing the Lifelong Head to monitor progress without needing to physically visit each office to follow up. The system also ensures that each requestor's uploaded documents remain private, as Google Drive folders are only accessible to the individual requestor and the Lifelong Head, protecting the confidentiality of personal training records.~~

Teaching and Non-Teaching Personnel. Requestors can prepare a training request and submit supporting documents through the system before completing any physical steps that the approving offices still require. They can then view the request's recorded status timeline rather than relying only on informal follow-ups. LEAPRS limits access to application records by role and department and stores file attachments in Google Drive. Direct sharing permissions for Drive folders should be checked separately before any claim is made that only the requestor and the Lifelong Learning Head can open them.

Original line 47

The Researchers. The end-to-end development of LEAPRS spanning requirements analysis, system design, implementation, and user evaluation equips the research team with practical technical and managerial competencies that serve as a direct culmination of their BSIT program at PLP.

Original line 48

Future Researchers. This study provides a concrete reference point for those pursuing similar administrative system development within Philippine higher education, with opportunities to expand upon LEAPRS through mobile integration, government portal connectivity, or more advanced AI functionality.

Original line 49

~~Overall, LEAPRS represents a significant step toward modernizing the administrative operations of Pamantasan ng Lungsod ng Pasig. By addressing the inefficiencies of the current manual training requisition process through a web-based, AI-assisted platform, the system delivers practical and measurable benefits to all its intended users contributing not only to the efficiency of the university's capacity development program but also to the broader goal of digital transformation in Philippine higher education.~~

Overall, LEAPRS represents a proposed way to make PLP's training requisition information easier to submit, record, and review. Its web-based workflow and AI-assisted feedback summaries address specific administrative needs, while the study's evaluation will determine how well the system serves its intended users in practice.

Statement of the Problem

Original line 52

This study aims to develop and evaluate LEAPRS: An AI-Assisted Web-Based Training Requisition and Monitoring System for Pamantasan ng Lungsod ng Pasig.

Original line 53

Specifically, it seeks to answer the following questions:

Original line 54

1. What are the existing problems and limitations of the current manual training requisition process at Pamantasan ng Lungsod ng Pasig in terms of:

Original line 55

1.1 Document submission and routing

Original line 56

1.2 Budget monitoring and utilization tracking

Original line 57

1.3 Approval workflow management

Original line 58

1.4 Post-training document compliance

Original line 59

2. What features and functionalities should LEAPRS provide to improve the training requisition and monitoring process at Pamantasan ng Lungsod ng Pasig?

Original line 60

~~3. What thematic summaries and insights can LEAPRS generate from the satisfaction and post-seminar learning surveys to support the Lifelong Head's review of employee learnings and training effectiveness?~~

3. How can LEAPRS's participant and supervisor evaluation forms, rating summaries, and AI-assisted summaries of written feedback support the Lifelong Learning Head's review of completed training activities?

Original line 61

4. How do the intended users assess the LEAPRS based on its functionality, efficiency, compatibility, usability, reliability, security, maintainability, and portability?

Scope and Limitations of the Study

Original line 64

The development of LEAPRS for Pamantasan ng Lungsod ng Pasig will include:

Original line 65

~~-Training Requisition and Document Submission. Teaching and non-teaching personnel will submit their training requisition requests digitally and upload all required pre-requisition documents through the platform.~~

- Training Requisition and Document Submission. Teaching and non-teaching personnel will submit requests associated with a CapDev project and upload the supporting documents required by the configured form.

Original line 66

~~-Approval Routing and Status Monitoring. Submitted requests will be routed automatically through the designated approval workflow, with both the Lifelong Head and the requesting personnel able to monitor application progress at any stage.~~

- Request Status Monitoring. Authorized users will record developments in each request's status timeline. A request can be stopped, resumed, completed, or denied through the appropriate actions. The system records these developments but does not automatically route physical documents among approving offices.

Original line 67

~~-Email Notification. The system sends email notification to Lifelong Learning Head whenever there is a submitted request and update to the status of a training requisition submitted by a requestor, allowing the Lifelong Learning Head to stay informed of requisition and to the progress without needing to manually check each request or physically follow up with the concerned office.~~

- In-App Notification. LEAPRS will show relevant request and status activity to involved users according to their role and department, allowing the Lifelong Learning Head to notice recorded changes without routinely asking each office for an update.

Original line 68

~~-Budget Monitoring and Utilization Tracking. Training budgets encoded from the CAPDEV will be tracked per activity, providing the Lifelong Head with a continuous and accurate view of fund utilization throughout the school year.~~

- Budget Monitoring and Utilization Tracking. CapDev projects will store an initial and remaining budget. LEAPRS checks whether a requested amount is within the remaining balance and records an

authorized deduction when the relevant status action is saved.

Original line 69

~~-Post-Training Document Compliance. Requestors will upload post-training documents, including attendance files, upon activity completion. The system monitors submission completeness and flags any missing documents accordingly.~~

- Post-Training Document Review. Requestors and authorized staff can attach files to configured request fields and status updates, allowing submitted documents to be reviewed with the request history. The system does not automatically determine whether all required post-training documents are complete.

Original line 70

~~-Audit Trail. All requisition actions, approval decisions, and system transactions are recorded and stored, giving the Lifelong Learning Head a transparent and traceable log of all program activities.~~

- Audit Trail. LEAPRS records user, CapDev, request, status-update, form-configuration, and maintenance actions in a searchable audit history available to Admins. Audit records remain available even if their related business records are deleted.

Original line 74

~~-AI-Assisted Post-Seminar Evaluation Analysis. The system will process responses from the post-seminar satisfaction survey and post-seminar learning survey submitted by requestors after each training activity. Quantitative responses from the satisfaction survey will be computed into statistical summaries, while open-ended responses from the learning survey will be processed through the Gemini API to generate thematic summaries and surface notable responses. Both outputs will be combined into a single report, with a brief executive summary generated to support the Lifelong Learning Head's review of employee learnings and training effectiveness.~~

- AI-Assisted Post-Activity Evaluation Review. On completing a request, the system creates participant and supervisor evaluation forms. Rating responses are presented as distributions and averages. Written comments can be summarized with Groq into strengths, areas for improvement, and recommended actions. The reviewer can inspect the underlying responses alongside the generated summary.

Original line 72

~~-Google Drive Integration. Files uploaded by requestors through the system will be automatically saved to the Lifelong Learning Head's Google Drive account, where a dedicated folder will be created per requestor to organize all submitted documents systematically.~~

- Google Drive Integration. Request and status attachments will be stored in Google Drive, while LEAPRS will keep the returned file details in its database. Files are organized with a request-specific folder name containing the username, submission date, and request name.

Original line 73

~~-Excel Report Export. The Lifelong Learning Head can export requisition and budget utilization data from the admin dashboard as an Excel file, supporting offline record keeping and reporting needs outside the system.~~

- Excel Report Export. Authorized Admin and Viewer users can select CapDev projects and fields to generate an Excel monitoring workbook. Department Viewers see their department's records, while Admins and Viewers with all-department access have broader reporting visibility.

Original line 74

~~- Admin Self-Request. The Lifelong Learning Head can submit a training requisition on their own behalf directly through the admin dashboard, following the same requisition process applied to all teaching and non-teaching personnel.~~

- Admin Request Management. An authorized Admin can manage CapDev projects and requests in the shared portal. This supports the Lifelong Learning Head's oversight of training requisitions when the Head is assigned the appropriate application role.

Original line 75

~~- Role-Based Google Drive Access Control. The system implements role-based access control for all requestor-submitted documents stored in the Lifelong Learning Head's Google Drive. Each requestor's dedicated folder is only accessible to that individual requestor and the Lifelong Learning Head, ensuring that personal training records remain confidential and protected from unauthorized access.~~

- Role-Based Access Control. LEAPRS controls visibility and actions in the application according to user role and department. This protection applies to portal records and actions; direct Google Drive folder access must be evaluated separately.

Original line 76

~~- Dashboard Data Visualization. The admin dashboard will display the following visual reports to support monitoring and decision-making: Summary Cards, Budget Utilization Gauge, Training Type Distribution, Activity Timeline, Seminar Attendance, Submission Status, and Per Faculty Office.~~

- Dashboard Data Visualization. Analytics will present operational views of CapDev budgets, requests, and status activity, including balance utilization, allocation by department, and training type distribution. The monthly Audit log activity chart currently uses sample values and will not be treated as a live measure in this study.

Original line 77

This study is subject to the following limitations:

Original line 78

~~- Mobile Application. A dedicated mobile application is not included in this study. However, LEAPRS will remain fully accessible through any web browser on any device.~~

- Mobile Application. A dedicated mobile application is not included in this study. LEAPRS is a browser-based system; its use on different screen sizes should be assessed as part of the system evaluation.

Original line 79

- Government Portal Integration. Integration with external government portals such as the Department of Budget and Management (DBM) or Commission on Audit (COA) systems is not covered in this study.

Original line 80

~~- AI Data Scope. The AI-assisted feature is limited to processing free-text responses submitted through the Post-Seminar Learning Survey (Form 2) within LEAPRS. It does not retrieve or reference information from outside the system, and its output is limited to thematic summaries and notable-response identification based solely on the responses submitted for a given seminar.~~

- AI Data Scope. AI assistance in LEAPRS includes a help chat and activity-design request drafting through Gemini, as well as Groq-generated summaries of written evaluation feedback when configured. The study does not include clustering-based thematic analysis or automatic judgments about training effectiveness. A user reviews an extracted request draft before confirming its submission, and evaluation summaries remain aids to human interpretation.

Original line 81

- Respondent Coverage. LEAPRS is designed to support the entire training requisition process of Pamantasan ng Lungsod ng Pasig, but the evaluation of the system in this study is limited to selected teaching personnel from the College of Computer Studies, the initial pilot department for teaching personnel, along with selected non-teaching personnel from PLP's administrative offices and the Lifelong Learning Head. This allows the system to be tested and refined with a focused group representative of its primary user types before wider deployment across the university. Full validation across all colleges and offices will be part of future rollout beyond the scope of this study.

Source verification notes

The works below were checked against publisher, institutional, journal, or author records. These notes support citation checking and are not part of the Chapter 1 prose. Full reference formatting should follow the format required by your adviser.

1. **Wibawa (2024)**. Digital Transformation and Administrative Efficiency Study. <https://doi.org/10.51699/ajslid.v3i1.3444>
2. **Kayanja, Kyambade, and Kiggundu (2025)**. Exploring digital transformation in higher education setting. <https://doi.org/10.1080/2331186X.2025.2489800>
3. **Fedeli, Liotino, and Taylor (2025)**. New Faculty Development: The Case of University of Padova. <https://doi.org/10.1002/ace.70006>
4. **Alonzo (2025)**. Bridging skills gaps: A learning needs analysis of non-teaching personnel. <https://www.ijriss.com/index.php/ijriss/article/download/4569/669/7416>
5. **Piloton et al. (2023)**. Development of Records Tracking Management System with QR Code. <https://www.ijfmr.com/publication-archive.php?issue=4&volume=5>
6. **Widiyatmoko, Nugroho, and Wiyanto (2024)**. Development of Web-Based Student Registration Information System. <https://doi.org/10.47709/cnahpc.v6i1.3459>
7. **Afandi and Nuralmasari (2026)**. Web-Based Digital Document Governance System. <https://ejournal.nusamandiri.ac.id/journal/index.php/pilar/article/download/8284/1681/32078>
8. **Pajenado and Rivera (2025)**. Digitalizing Government Services Through the Paspas Permit Implementation. <https://doi.org/10.11594/ijmaber.06.04.11>
9. **De Paoli (2024)**. Further explorations on the use of large language models for thematic analysis. <https://doi.org/10.17169/fqs-25.3.4196>
10. **Wongvorachan and Bulut (2026)**. The Use of Natural Language Processing in Learning Analytics. https://doi.org/10.1007/978-3-031-95365-1_9
11. **Katz, Coloyan Fleming, and Main (2026)**. Thematic analysis with open-source generative AI and machine learning. <https://doi.org/10.1057/s41599-026-06508-5>

Citation corrections: Wongvorachan and Bulut's repository record lists 2025, but the published book chapter citation is 2026. Katz et al. circulated a 2024 preprint; the journal article is dated 2026. The former Madyatmadja et al. (2022) dashboard claim was removed because the described source could not be matched reliably. The DBSCAN/HDBSCAN/OPTICS discussion was removed because LEAPRS does not implement that pipeline.